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Physics-informed modeling of salinity-gradient power generation and operating conditions

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Physics-informed modeling of salinity-gradient power generation and operating conditions

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Data availability

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

22 July 2026

Dear Editor,

Please consider our manuscript, “Physics-informed modeling of salinity-gradient power generation and operating conditions,” for publication as an Original Research Article in ***Renewable Energy***.

The manuscript examines how hydraulic pressure and inlet salinities shape membrane-scale power generation from salinity gradients. The feed and draw channels are modeled separately and coupled through the membrane transport conditions. The model is evaluated using 210 finite-element cases, independent numerical checks, five training runs, unseen operating points, and comparisons with global physics-informed and operator-learning models.

The results recover the pressure-dependent power maximum and the increase in available power with salinity difference. Mean gross-power error is 2.40%, and the fitted finite-element and surrogate optimal-pressure maps differ by 0.03 bar on average. Full-field prediction is 175 times faster, and the combined offline cost is recovered after about 504 operating queries.

The resulting pressure-salinity maps support rapid screening before selected conditions are examined with high-fidelity simulation. The reported power values are limited to membrane-scale gross power.

The manuscript is original, has been approved by all authors, and is submitted exclusively to ***Renewable Energy***. The authors declare no competing interests.

Thank you for your consideration.

Sincerely,

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Highlights

- Pressure-salinity maps identify favorable salinity-gradient power conditions.
- Mean gross-power error across five runs is 2.40%.
- Fitted optimal-pressure maps differ by 0.03 bar on average.
- Membrane-region error is 20.3% lower than in a matched global model.
- Full-field prediction is 175 times faster; break-even occurs at 504 queries.

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Abstract

Salinity-gradient power generation converts the energy released by mixing low- and high-salinity water into hydraulic power. Recoverable gross power depends on the applied pressure and the salinity difference across the membrane, so a search for suitable operating conditions can require many transport simulations. A physics-informed model represents the feed and draw channels separately and couples them through membrane transport conditions. The numerical analysis used 210 finite-element simulations spanning the main operating range, unseen points, and nearby boundary conditions. Mean errors were $2.54 \pm 0.17\%$ for concentration, $2.40 \pm 0.19\%$ for gross power, $2.82 \pm 0.11\%$ near the membrane, and $1.12 \pm 0.05\%$ for interface flux balance. The model reproduced the pressure-dependent power maximum and the increase in available power with salinity difference. Smooth operating maps fitted to the held-out results differed by 0.03 bar on average for optimal pressure and by 0.48% for maximum gross power. A full-field prediction required 0.087 s, giving a 175-fold speed increase. The combined data-generation and training cost was offset after about 504 queries. The model enables rapid screening of pressure and salinity combinations before selected conditions are examined with high-fidelity simulation.

Keywords: Pressure-Retarded Osmosis; Salinity-Gradient Energy; Gross Power Density; Physics-Informed Modeling; Operating Conditions; Surrogate Modeling

1. Introduction

Salinity-gradient energy can be harvested wherever water streams with contrasting salt concentrations coexist. Pressure-Retarded Osmosis (PRO) converts this gradient into hydraulic work by drawing water from a low-salinity feed into a pressurized draw stream. For a given feed-draw pair,

6 hydraulic pressure is the central operating decision. Low pressure leaves part
7 of the available mixing energy unused, and excessive pressure suppresses water
8 transfer. Studies in *Renewable Energy* have examined full-scale operating
9 optimization [1], spiral-wound module behavior [2], alternative module layouts
10 [3], and non-ideal process models [4]. Across these studies, pressure, salinity,
11 flow rate, and membrane transport define the useful operating range.

12 Gross areal power equals hydraulic pressure multiplied by membrane-
13 averaged water flux. Early PRO experiments established a pressure-dependent
14 maximum [5], and later analysis linked its location to the membrane-side
15 osmotic-pressure difference [6]. Concentration polarization changes the lo-
16 cal driving force, as summarized in the feasibility review by Lee et al. [7].
17 Membrane-focused work further shows how material properties and polariza-
18 tion limit performance [8]. Thermodynamic and module-scale losses shape
19 the pressure–power response [9], and finite salt permeability adds another per-
20 formance limit [10]. Commercial-length channel models resolve axial changes
21 in concentration and flow [11]. Module-scale analysis captures the resulting
22 variation along the membrane [12], and spiral-wound models quantify the
23 additional effects of channel configuration and pressure loss [13]. The scalar
24 power response depends on the membrane-side concentration field.

25 High-fidelity transport models resolve channel flow, ionic transport, elec-
26 trostatic closure, membrane-normal water transfer, and the concentration
27 gradients entering the power calculation. Recent reviews trace the devel-
28 opment from analytical descriptions to detailed numerical and multiscale
29 formulations [14]. These models are practical for selected conditions. Dense
30 pressure–salinity maps are useful for comparing water-source combinations
31 and locating the pressure range that yields the highest gross power. Pro-
32 ducing such maps may demand hundreds or thousands of forward solutions,
33 making repeated evaluation a central cost in full-scale optimization [1] and
34 process-design studies [4].

35 Reduced-order and data-driven surrogates lower online cost by learning
36 a parameter-to-field map from completed simulations. Proper Orthogonal
37 Decomposition with Radial Basis Function Interpolation (POD–RBF) is
38 efficient for low-dimensional solution manifolds [15]. Deep Operator Network
39 (DeepONet) learns nonlinear operators from paired functions and responses
40 [16]. Fourier Neural Operator (FNO) represents parametric solution maps
41 through spectral convolutions [17]; the broader neural-operator framework
42 is reviewed by Kovachki et al. [18]. Physics-informed operator learning adds
43 equation residuals to the operator objective [19]. Their accuracy depends on

the coverage of the training fields. The membrane occupies a narrow part of the domain, so a small global error can coexist with a larger interface error that directly affects water flux and power.

Physics-Informed Neural Networks (PINNs) combine sparse observations with residuals of the governing equations and boundary conditions [20]. Their engineering and multiphysics applications are reviewed by Karniadakis et al. [21] and Cuomo et al. [22]; mixed-variable formulations extend the approach to coupled fields [23]. A parameterized network can span a family of operating conditions. Training can become unstable when residual terms generate gradients of very different magnitudes [24]. Standard training also becomes sensitive as the governing problem and parameter range grow more difficult [25]. Unequal convergence rates among loss components have been linked to training failure [26], and broad benchmarks confirm strong sensitivity to formulation and optimization choices across partial differential equations [27]. Residual connections support the optimization of deeper networks [28]. Trainable activation slopes [29], adaptive weighting [30], and residual-based sampling [31] address distinct parts of the training problem. In PRO, a single global network must also resolve the concentration contrast across the membrane.

Domain decomposition assigns separate approximators to physically distinct regions. The Extended Physics-Informed Neural Network (XPINN) couples space–time subdomains through interface residuals [32]. Conservative Physics-Informed Neural Networks impose conservation across discrete subdomains [33]. Finite-Basis Physics-Informed Neural Networks (FBPINNs) localize the approximation with overlapping windows [34]. A gating network allows adaptive soft decomposition [35], and multilevel Schwarz-type architectures extend the idea across scales [36]. PRO has a natural two-domain geometry, with one network for the feed channel and one for the draw channel. Water permeation, normal-velocity consistency, tangential no-slip, solute rejection, and electrical insulation couple the two predictions at the membrane. The arrangement preserves distinct channel states and applies interface residuals directly to the quantities used in the gross-power calculation.

To separate architectural effects from network size, the comparison includes a parameter-matched global Residual-Network Physics-Informed Neural Network (ResNet-PINN). Operator models receive complete training fields, and the PINNs use sparse Finite Element Method (FEM) anchors with physics residuals. Each finite-element case belongs wholly to the training, validation,

test, or boundary and Out-of-Distribution (OOD) partitions. Spatial samples from the same reference field never cross partitions.

Evaluation covers global concentration error, membrane-band error, interface-flux imbalance, mass closure, gross-power error, condition-level variation, boundary behavior, optimal-pressure agreement, and computational break-even. Independent quadratic fits to the held-out finite-element and surrogate outputs summarize pressure–salinity trends. Condition-wise surrogate errors are calculated directly from the original held-out cases. The framework recovers the pressure-dependent gross-power response across feed and draw salinities and retains the membrane-scale fields that govern it.

The study combines four elements:

1. pressure–salinity maps for recovering preferred pressure and maximum gross power across the modeled operating range;
2. a parameterized membrane-scale model linking hydraulic pressure and inlet salinities to coupled transport fields, membrane water flux, and gross power density;
3. a two-subdomain ResNet-PINN with direct membrane coupling and sparse finite-element anchors drawn exclusively from training cases; and
4. a capacity-controlled comparison with global PINNs, DeepONet, FNO, and POD–RBF, together with condition-level analysis, online timing, and end-to-end query cost.

2. Salinity-gradient power transport and gross-power model

2.1. Geometry, variables, and scope

We consider a steady two-dimensional domain

$$\Omega = [0, L] \times [0, H], \quad (1)$$

with an internal membrane at

$$x = L_m, \quad 0 < L_m < L. \quad (2)$$

The draw channel occupies $0 \leq x < L_m$ and the feed channel occupies $L_m < x \leq L$, defining the subdomains $\Omega^{(d)}$ and $\Omega^{(f)}$. The coordinate x is normal to the membrane and y is tangential to it. Inlet and outlet boundaries are located at $y = 0$ and $y = H$; $x = 0$ and $x = L$ define the outer

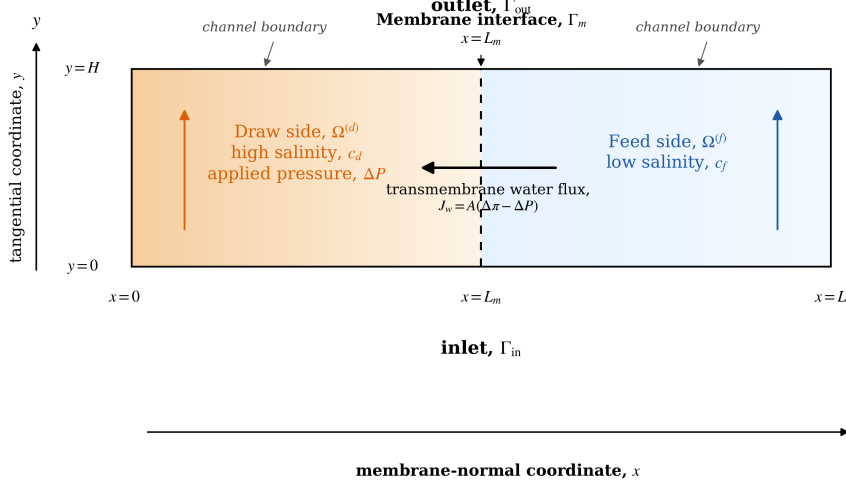


Figure 1: Membrane-scale PRO domain.

channel walls. Figure 1 shows the reduced geometry, including the channel arrangement and coordinate directions.

The state variables are velocity $\mathbf{u} = (u_x, u_y)$, pressure p , ionic concentrations c_i for $i \in \{+, -\}$, and electric potential ϕ . The fluid is Newtonian, incompressible, and isothermal, with constant density, viscosity, diffusivities, valences, and permittivity. The membrane is a zero-thickness interface with complete salt rejection. The model excludes support-layer transport, reverse salt flux, spacers, fouling, concentration-dependent properties, and module-scale pressure loss.

Mesh resolution and collocation density are increased near Γ_m , where the concentration changes most sharply and the membrane-normal velocity enters the power calculation.

For a 1:1 electrolyte, the osmotic pressure is approximated by the van't Hoff relation

$$\pi = RT \sum_i c_i, \quad (3)$$

where concentrations in the governing equations are expressed in mol m^{-3} . Molar operating values are converted before Eq. (3) is evaluated. This ideal-solution form is widely used in PRO analysis [6]. Non-ideal seawater properties

128 become more important at high salinity [37], as discussed in recent PRO
129 reviews [14].

130 2.2. Governing equations

131 2.2.1. Incompressible flow

132 In each subdomain, the steady incompressible Navier–Stokes equations
133 are

$$\nabla \cdot \mathbf{u} = 0, \quad (4)$$

$$\rho(\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \mu \nabla^2 \mathbf{u}. \quad (5)$$

135 The Cauchy stress is

$$\boldsymbol{\sigma} = -p \mathbf{I} + \mu (\nabla \mathbf{u} + \nabla \mathbf{u}^\top). \quad (6)$$

136 2.2.2. Ion transport

137 For each ionic species,

$$\nabla \cdot \mathbf{J}_i = 0, \quad (7)$$

138 with Nernst–Planck flux

$$\mathbf{J}_i = \mathbf{u} c_i - D_i \nabla c_i - \frac{z_i D_i F}{RT} c_i \nabla \phi, \quad i \in \{+, -\}. \quad (8)$$

139 Equivalently,

$$\nabla \cdot \left(\mathbf{u} c_i - D_i \nabla c_i - \frac{z_i D_i F}{RT} c_i \nabla \phi \right) = 0. \quad (9)$$

140 2.2.3. Electrostatic closure

141 The auxiliary potential field satisfies

$$-\nabla \cdot (\varepsilon \nabla \phi) = \rho_e, \quad \rho_e = F \sum_i z_i c_i, \quad (10)$$

142 where $\varepsilon = \varepsilon_r \varepsilon_0$. The equation is used as a bulk electrostatic closure. Interfacial
143 diffuse-charge dynamics are outside the resolved length scale of the model
144 [38].

145 *2.3. External boundary conditions*

146 Let $\Gamma_{\text{in}}^{(s)}$, $\Gamma_{\text{out}}^{(s)}$, and $\Gamma_w^{(s)}$ denote the inlet, outlet, and outer wall of subdomain
 147 $s \in \{f, d\}$, and let $\mathbf{n}^{(s)}$ be the outward unit normal on its external boundary.
 148 A prescribed laminar tangential velocity profile is imposed at each inlet,

$$\mathbf{u}^{(s)} = \mathbf{u}_{\text{in}}(x) \quad \text{on } \Gamma_{\text{in}}^{(s)}, \quad (11)$$

149 and the outer walls are no-slip,

$$\mathbf{u}^{(s)} = 0 \quad \text{on } \Gamma_w^{(s)}. \quad (12)$$

150 The hydraulic pressure difference enters the flow problem through side-
 151 specific outlet tractions. The feed outlet defines the pressure datum, and the
 152 draw outlet carries the imposed pressure ΔP :

$$\boldsymbol{\sigma}^{(f)} \mathbf{n}^{(f)} = 0 \quad \text{on } \Gamma_{\text{out}}^{(f)}, \quad (13)$$

$$\boldsymbol{\sigma}^{(d)} \mathbf{n}^{(d)} = -\Delta P \mathbf{n}^{(d)} \quad \text{on } \Gamma_{\text{out}}^{(d)}. \quad (14)$$

153 This definition gives $\Delta P = p_d - p_f > 0$ for a pressurized draw channel. The
 154 membrane supports this pressure difference, and pressure and normal traction
 155 remain discontinuous across the zero-thickness interface.

156 At the inlets,

$$c_i^{(s)} = c_{i,\text{in}}^{(s)} \quad \text{on } \Gamma_{\text{in}}^{(s)}. \quad (15)$$

157 Outer walls are impermeable to ions,

$$\mathbf{J}_i^{(s)} \cdot \mathbf{n}^{(s)} = 0 \quad \text{on } \Gamma_w^{(s)}, \quad (16)$$

158 and the diffusive and migrative outlet components vanish,

$$\left(-D_i \nabla c_i^{(s)} - \frac{z_i D_i F}{RT} c_i^{(s)} \nabla \phi^{(s)} \right) \cdot \mathbf{n}^{(s)} = 0 \quad \text{on } \Gamma_{\text{out}}^{(s)}. \quad (17)$$

159 Electrically insulating conditions are applied at the outer walls and outlets,

$$\nabla \phi^{(s)} \cdot \mathbf{n}^{(s)} = 0 \quad \text{on } \Gamma_w^{(s)} \cup \Gamma_{\text{out}}^{(s)}. \quad (18)$$

160 Because the ideal rejecting membrane electrically separates the subdomains,
 161 each potential field has its own gauge condition,

$$\phi^{(s)} = 0 \quad \text{on } \Gamma_{\text{ref}}^{(s)} \subseteq \Gamma_{\text{in}}^{(s)}. \quad (19)$$

162 *2.4. Membrane-interface closure*

163 Let $\mathbf{m}^{(d)} = -\mathbf{e}_x$ and $\mathbf{m}^{(f)} = \mathbf{e}_x$ denote unit normals directed from the
 164 membrane into the draw and feed channels, respectively, and let $\mathbf{t}$ be the
 165 tangential unit vector. The membrane-side osmotic-pressure difference is

$$\Delta\pi(y) = RT \sum_i \left[c_i^{(d)}(L_m^-, y) - c_i^{(f)}(L_m^+, y) \right]. \quad (20)$$

166 The local water flux is

$$J_w(y) = A [\Delta\pi(y) - \Delta P]. \quad (21)$$

167 Kinematic consistency requires

$$\mathbf{u}^{(d)} \cdot \mathbf{m}^{(d)} = J_w, \quad \mathbf{u}^{(f)} \cdot \mathbf{m}^{(f)} = -J_w \quad \text{on } \Gamma_m. \quad (22)$$

168 The stationary membrane imposes tangential no-slip on both faces,

$$\mathbf{u}^{(f)} \cdot \mathbf{t} = 0, \quad \mathbf{u}^{(d)} \cdot \mathbf{t} = 0 \quad \text{on } \Gamma_m. \quad (23)$$

169 For the perfectly rejecting baseline membrane,

$$\mathbf{J}_i^{(f)} \cdot \mathbf{m}^{(f)} = 0, \quad \mathbf{J}_i^{(d)} \cdot \mathbf{m}^{(d)} = 0 \quad \text{on } \Gamma_m, \quad i \in \{+, -\}, \quad (24)$$

170 and no normal electrical current is admitted,

$$\nabla\phi^{(f)} \cdot \mathbf{m}^{(f)} = 0, \quad \nabla\phi^{(d)} \cdot \mathbf{m}^{(d)} = 0 \quad \text{on } \Gamma_m. \quad (25)$$

171 Equations (13)–(25) close the two fluid subproblems and preserve the pressure
 172 and potential discontinuities of the ideal membrane.

173 *2.5. Gross power density and operating variables*

174 The membrane-averaged water flux and areal power density are

$$\bar{J}_w = \frac{1}{|\Gamma_m|} \int_{\Gamma_m} J_w \, ds, \quad P = \bar{J}_w \Delta P. \quad (26)$$

175 The formulation retains the membrane-side concentration gradients that de-
 176 termine water flux and gross power density. The three operating inputs are
 177 hydraulic pressure, feed salinity, and draw salinity. Positive water transfer
 178 requires $\Delta P < \Delta\pi$ locally. If the membrane-side osmotic jump were indepen-
 179 dent of pressure, substitution of Eq. (21) into Eq. (26) would give a concave
 180 quadratic power response with an ideal maximum at $\Delta P = \Delta\pi/2$. In the
 181 coupled model, $\Delta\pi$ varies along the membrane and changes with the flow
 182 and concentration fields, so the maximizing pressure must be recovered from
 183 the full transport solution. The resulting surrogate is used to evaluate this
 184 membrane-scale operating window over pressure and salinity.

185 3. Membrane-aligned physics-informed surrogate

186 3.1. Domain-decomposed representation

187 Two subnetworks approximate the feed and draw solutions,

$$\mathcal{N}_{\theta_f}^{(f)} : (x, y, \Delta P, c_f, c_d) \mapsto \left(u_x^{(f)}, u_y^{(f)}, p^{(f)}, c_+^{(f)}, c_-^{(f)}, \phi^{(f)} \right), \quad (27)$$

$$\mathcal{N}_{\theta_d}^{(d)} : (x, y, \Delta P, c_f, c_d) \mapsto \left(u_x^{(d)}, u_y^{(d)}, p^{(d)}, c_+^{(d)}, c_-^{(d)}, \phi^{(d)} \right). \quad (28)$$

188 The networks share an architecture and have independent parameters. Hy-
 189 draulic pressure and inlet concentrations enter both subnetworks, allowing
 190 one trained model to cover the full parameter window. Each interior point
 191 is routed to its physical channel; membrane points pass through both sub-
 192 networks. Paired feed-side and draw-side predictions at the same tangential
 193 coordinate define the coupling loss. Separate interface values are retained on
 194 the two sides, preserving the concentration jump used by the water-flux law.

195 Each subnetwork uses fully connected residual blocks,

$$\mathbf{h}^{(\ell+1)} = \mathbf{h}^{(\ell)} + \mathcal{F}(\mathbf{h}^{(\ell)}; W^{(\ell)}, b^{(\ell)}), \quad (29)$$

196 with trainable activation scaling

$$\sigma_a(z) = \tanh(az). \quad (30)$$

197 Residual connections support optimization under the composite objective [28].
 198 The learned slope a follows the adaptive-activation formulation of Jagtap
 199 et al. [29] and adjusts the effective nonlinearity across the smooth channel
 200 interiors and the sharper membrane-adjacent region.

201 3.2. Normalization

202 Coordinates and operating variables are mapped to $[-1, 1]$ using the limits
 203 of the in-domain training box. For example,

$$x' = 2\frac{x}{L} - 1, \quad y' = 2\frac{y}{H} - 1, \quad \Delta P' = 2\frac{\Delta P - \Delta P_{\min}}{\Delta P_{\max} - \Delta P_{\min}} - 1, \quad (31)$$

204 with analogous transformations for c_f and c_d . Outputs are scaled by $U_0 =$
 205 10^{-3} m s^{-1} , $p_0 = 10^6 \text{ Pa}$, $c_0 = 10^3 \text{ mol m}^{-3}$, and $\phi_0 = RT/F$:

$$\tilde{\mathbf{u}} = \frac{\mathbf{u}}{U_0}, \quad \tilde{p} = \frac{p}{p_0}, \quad \tilde{c}_i = \frac{c_i}{c_0}, \quad \tilde{\phi} = \frac{\phi}{\phi_0}. \quad (32)$$

206 All Partial Differential Equation (PDE) residuals and interface residuals are
 207 evaluated in nondimensional form before loss aggregation.

208 *3.3. Physics-informed objective*

209 Let $R_{\text{cont}}^{(s)}$, $R_{\text{mom}}^{(s)}$, $R_i^{(s)}$, and $R_\phi^{(s)}$ denote the nondimensional residuals of
 210 Eqs. (4), (5), (9), and (10). The subdomain PDE loss is

$$\mathcal{L}_{\text{PDE}}^{(s)} = \lambda_{\text{cont}} \langle |R_{\text{cont}}^{(s)}|^2 \rangle + \lambda_{\text{mom}} \langle \|R_{\text{mom}}^{(s)}\|_2^2 \rangle + \lambda_{\text{sp}} \sum_i \langle |R_i^{(s)}|^2 \rangle + \lambda_\phi \langle |R_\phi^{(s)}|^2 \rangle, \quad (33)$$

211 where $\langle \cdot \rangle$ denotes the sample mean. The total PDE term is

$$\mathcal{L}_{\text{PDE}} = \mathcal{L}_{\text{PDE}}^{(f)} + \mathcal{L}_{\text{PDE}}^{(d)}. \quad (34)$$

212 The Boundary Condition (BC) loss collects the inlet, outer-wall, side-specific
 213 outlet-traction, species, and potential conditions:

$$\begin{aligned} \mathcal{L}_{\text{BC}} = & \lambda_{u,\text{in}} \mathcal{L}_{u,\text{in}} + \lambda_{u,w} \mathcal{L}_{u,w} + \lambda_{\sigma,\text{out}} \mathcal{L}_{\sigma,\text{out}} + \lambda_{c,\text{in}} \mathcal{L}_{c,\text{in}} \\ & + \lambda_{\text{sp},w} \mathcal{L}_{\text{sp},w} + \lambda_{c,\text{out}} \mathcal{L}_{c,\text{out}} + \lambda_{\phi,\text{neu}} \mathcal{L}_{\phi,\text{neu}} + \lambda_{\phi,\text{ref}} \mathcal{L}_{\phi,\text{ref}}. \end{aligned} \quad (35)$$

214 At the membrane, the Interface (IF) residuals are

$$R_{J_w} = J_w - A(\Delta\pi - \Delta P), \quad (36)$$

$$R_u^{(d)} = \mathbf{u}^{(d)} \cdot \mathbf{m}^{(d)} - J_w, \quad R_u^{(f)} = \mathbf{u}^{(f)} \cdot \mathbf{m}^{(f)} + J_w, \quad (37)$$

$$R_t^{(s)} = \mathbf{u}^{(s)} \cdot \mathbf{t}, \quad (38)$$

$$R_{i,m}^{(s)} = \mathbf{J}_i^{(s)} \cdot \mathbf{m}^{(s)}, \quad (39)$$

$$R_{\phi,m}^{(s)} = \nabla \phi^{(s)} \cdot \mathbf{m}^{(s)}. \quad (40)$$

215 The interface loss is

$$\begin{aligned} \mathcal{L}_{\text{IF}} = & \lambda_{J_w} \langle |R_{J_w}|^2 \rangle + \lambda_n \sum_s \langle |R_u^{(s)}|^2 \rangle + \lambda_t \sum_s \langle |R_t^{(s)}|^2 \rangle \\ & + \lambda_{\text{sol}} \sum_{s,i} \langle |R_{i,m}^{(s)}|^2 \rangle + \lambda_{\phi,m} \sum_s \langle |R_{\phi,m}^{(s)}|^2 \rangle. \end{aligned} \quad (41)$$

216 The complete objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{PDE}} + \mathcal{L}_{\text{BC}} + \mathcal{L}_{\text{IF}} + \lambda_{\text{data}} \mathcal{L}_{\text{data}}. \quad (42)$$

217 The supervised term uses sparse finite-element anchors drawn only from
 218 training conditions. Anchors are distributed across the two channels and
 219 contain the six reference outputs at selected mesh nodes. They stabilize the
 220 parameter-to-field map, and the PDE, boundary, and interface terms govern
 221 the unsupervised locations. The interface residual is evaluated independently
 222 of the anchor set, and membrane conservation follows from the paired interface
 223 conditions.

224 3.4. Adaptive weighting, sampling, and optimization

225 For loss component j , the weight is updated from its parameter-gradient
226 norm,

$$\lambda_j^{(t+1)} = (1 - \beta)\lambda_j^{(t)} + \beta \frac{\bar{g}^{(t)}}{g_j^{(t)} + \epsilon}, \quad g_j^{(t)} = \|\nabla_{\theta} \mathcal{L}_j^{(t)}\|_2. \quad (43)$$

227 Here, $\bar{g}^{(t)}$ is the mean gradient norm across the active loss components.
228 Weights are updated every 200 Adam iterations. Because the residuals are
229 nondimensionalized, equal coefficients are used in the sampling indicator

$$E(\mathbf{x}) = |R_{\text{cont}}| + \|R_{\text{mom}}\|_2 + \sum_i |R_i| + |R_{\phi}|. \quad (44)$$

230 Every 500 Adam iterations, 1,000 high-residual points are selected from a
231 5,000-point candidate pool and combined with 300 additional membrane-band
232 points. The retained high-residual set changes during training, allowing the
233 collocation distribution to follow the unresolved parts of the pressure–salinity
234 family. The fixed membrane-band enrichment prevents the adaptive pool from
235 becoming dominated by the larger channel interiors. Training uses 20,000
236 Adam iterations followed by Limited-Memory Broyden–Fletcher–Goldfarb–
237 Shanno (L-BFGS) refinement with validation-based early stopping. Validation
238 loss and validation power error are monitored during model selection; test and
239 OOD cases remain untouched. The timed five-seed runs terminate between
240 23,124 and 31,564 total iterations.

241 4. Numerical design and evaluation protocol

242 4.1. Reference solutions and operating conditions

243 COMSOL provides the primary finite-element solutions, and the FEniCS
244 finite-element framework supplies an independent numerical check. Both
245 implementations use the geometry, governing equations, side-specific pressure
246 conditions, and membrane closure defined above. Membrane-refined coarse,
247 medium, and fine meshes contain approximately 5.21×10^4 , 1.52×10^5 ,
248 and 3.48×10^5 degrees of freedom. Refinement is concentrated around the
249 membrane and where concentration gradients change most rapidly. Across
250 six grid-check conditions, the medium-to-fine differences average 0.371% for
251 water flux, 0.415% for power density, and 0.704% for membrane concentration;
252 the largest difference is 0.937%. The medium mesh is used for the parametric

Table 1: Physical parameters.

Parameter	Description	Value
L	Membrane-normal extent	5×10^{-3} m
H	Tangential extent	5×10^{-4} m
L_m	Membrane location	2.5×10^{-3} m
ρ	Fluid density	1000 kg m ⁻³
μ	Dynamic viscosity	1.0×10^{-3} Pa s
D_+	Na ⁺ diffusivity	1.33×10^{-9} m ² s ⁻¹
D_-	Cl ⁻ diffusivity	2.03×10^{-9} m ² s ⁻¹
z_+, z_-	Ionic valences	+1, -1
ε_r	Relative permittivity	78.5
A	Effective water permeability	4.5×10^{-12} m Pa ⁻¹ s ⁻¹
T	Temperature	298 K
u_{in}	Mean inlet velocity	1.0×10^{-3} m s ⁻¹

study. Its discretization differences remain below the surrogate errors, and its solution time permits evaluation of the complete operating set.

The in-domain parameter box is

$$\Delta P \in [4, 16] \text{ bar}, \quad c_f \in [0, 0.08] \text{ mol L}^{-1}, \quad c_d \in [0.45, 0.80] \text{ mol L}^{-1}. \quad (45)$$

It contains 180 distinct conditions: 126 for training, 27 for validation, and 27 for testing. Thirty additional boundary/OOD cases extend pressure to $[2, 4] \cup [16, 18]$ bar, feed concentration to 0.10 mol L^{-1} , and draw concentration to $[0.35, 0.45] \cup [0.80, 0.90] \text{ mol L}^{-1}$. These cases include single-boundary excursions and combined pressure-salinity extremes. Each finite-element solution belongs to a single partition. This condition-wise split prevents spatial information from a held-out reference field entering training.

4.2. Compared models and training protocol

Seven models are evaluated: Multilayer Perceptron Physics-Informed Neural Network (MLP-PINN), adaptive single ResNet-PINN, parameter-matched global ResNet-PINN, DeepONet, FNO, POD-RBF, and the membrane-aligned two-subdomain ResNet-PINN. The matched global network and the membrane-aligned model contain approximately 1.008×10^5 and 1.005×10^5 trainable parameters. PINN variants use finite-element anchors at 0.5% of

the nodes per training condition unless otherwise stated. DeepONet, FNO, and POD-RBF use the 126 complete training fields.

DeepONet uses a branch network for $(\Delta P, c_f, c_d)$ and a trunk network for (x, y) . FNO receives the operating parameters as spatially broadcast channels on the common output grid. POD-RBF interpolates the reduced coefficients needed to reconstruct the six output fields using Gaussian radial basis functions. DeepONet, FNO, and POD-RBF receive every nodal value from 126 complete fields. The PINNs receive 0.5% nodal anchors from the same cases together with collocation residuals. The comparison covers complete-field operator learning and sparsely supervised physics-informed learning. Hyperparameters are selected on the 27 validation conditions. Test and OOD conditions play no role in model selection.

Table 2: Model capacity and supervision.

Model	Parameters	Complete training fields	Anchors per condition	Physics residuals
MLP-PINN	34,000	0	0.5%	Yes
Adaptive single ResNet-PINN	51,000	0	0.5%	Yes
Matched global ResNet-PINN	100,800	0	0.5%	Yes
DeepONet	96,800	126	100%	No
FNO	108,200	126	100%	No
POD-RBF	18,200	126	100%	No
Two-subdomain ResNet-PINN	100,500	0	0.5%	Yes

The PINNs start with 10,000 interior, 1,500 external-boundary, and 500 membrane points before adaptive enrichment. The five seeds are 7, 21, 42, 77, and 123. Timing measurements use the same computational environment and include full-field reconstruction on the common evaluation grid. Since absolute wall-clock time depends on hardware, the analysis emphasizes within-environment speed ratios and break-even counts.

4.3. Metrics and statistical analysis

The global concentration error is the mean relative L_2 error of the two ionic concentrations,

$$E_c = \frac{100}{2} \sum_{i \in \{+, -\}} \frac{\|c_i - c_i^{\text{ref}}\|_2}{\|c_i^{\text{ref}}\|_2}. \quad (46)$$

The relative power-density error is

$$E_P = 100 \frac{|P - P^{\text{ref}}|}{|P^{\text{ref}}|}. \quad (47)$$

292 The membrane-band concentration error uses the same definition as Eq. (46),
 293 restricted to points within $0.05L$ of Γ_m . Interface-flux imbalance is

$$E_J = \frac{100}{2} \left[\frac{\|\mathbf{u}^{(d)} \cdot \mathbf{m}^{(d)} - J_w\|_2}{\|J_w\|_2} + \frac{\|\mathbf{u}^{(f)} \cdot \mathbf{m}^{(f)} + J_w\|_2}{\|J_w\|_2} \right]. \quad (48)$$

294 Combined-domain mass closure is evaluated on the external boundary,

$$E_M = 100 \frac{\left| \int_{\partial\Omega_{\text{ext}}} \mathbf{u} \cdot \mathbf{n} \, ds \right|}{\int_{\Gamma_{\text{in}}^{(f)} \cup \Gamma_{\text{in}}^{(d)}} |\mathbf{u} \cdot \mathbf{n}| \, ds}. \quad (49)$$

295 Internal membrane contributions cancel when the two subdomains are consid-
 296 ered together.

297 Seed-level results are reported as mean $\pm$ sample standard deviation.
 298 Condition-wise comparisons use two-sided Wilcoxon signed-rank tests. Holm
 299 correction is applied jointly to the 16 prespecified comparisons between the
 300 membrane-aligned model and the matched global PINN or FNO across four
 301 metrics and two evaluation partitions. Rank-biserial correlation reports paired
 302 effect size. Median, 90th-percentile, and maximum condition-wise errors are
 303 also examined for the membrane-aligned model. Pairwise win rates report the
 304 fraction of conditions on which it gives a lower error than the capacity-matched
 305 global PINN or FNO.

306 4.4. Operating-window post-processing

307 Performance metrics are calculated directly from held-out finite-element
 308 and network predictions. A separate response surface summarizes the en-
 309 ergy operating window. Let $\Delta c = c_d - c_f$. For the 57 in-domain test and
 310 boundary/OOD conditions, gross power is represented by

$$\hat{P} = b_0 + b_1 \Delta P + b_2 \Delta P^2 + b_3 \Delta c + b_4 \Delta c^2 + b_5 \Delta P \Delta c + b_6 c_f + b_7 \Delta P c_f. \quad (50)$$

311 The same form is fitted independently to the finite-element and membrane-
 312 aligned surrogate outputs. Five-fold cross-validation evaluates predictions
 313 on omitted folds and measures the accuracy of the smooth summary. Since
 314 $b_2 < 0$ for both fits, the pressure that maximizes fitted gross power within
 315 the in-domain interval is

$$\Delta P^* = \text{clip} \left[-\frac{b_1 + b_5 \Delta c + b_7 c_f}{2b_2}, 4, 16 \right] \text{ bar}. \quad (51)$$

316 The surfaces are fitted after the condition-wise evaluation and play no role in
 317 network training. Their cross-validated errors are reported separately from
 318 the neural-surrogate errors. The MLP-PINN curve in Fig. 4 uses the same
 319 fitted form for visual comparison and is excluded from the map statistics.
 320 Independent finite-element and membrane-aligned surrogate fits yield optimal-
 321 pressure and maximum-power maps over the same feed-draw grid. The map
 322 nodes evaluate the fitted surfaces; the finite-element data remain the original
 323 57 held-out conditions.

324 The operating consequence of choosing the surrogate optimum is summa-
 325 rized by the fitted-surface power regret

$$R_P = 100 \frac{\hat{P}_{\text{FEM}}(\Delta P_{\text{FEM}}^*) - \hat{P}_{\text{FEM}}(\Delta P_{\text{sur}}^*)}{\hat{P}_{\text{FEM}}(\Delta P_{\text{FEM}}^*)}. \quad (52)$$

326 The value is evaluated on the fitted finite-element surface at the pressure
 327 selected from the surrogate fit.

328 5. Results and discussion

329 5.1. Field accuracy and membrane-interface fidelity

330 Table 3 summarizes the five-seed comparison. All entries are mean $\pm$
 331 sample standard deviation, and all errors are percentages. Raising the capacity
 332 of the global PINN lowers mean concentration error from 3.21% for the
 333 adaptive single ResNet-PINN to 2.80% for the parameter-matched global
 334 network. Network size explains part of the gain over the smaller global model.
 335 The membrane-aligned split adds a further improvement under matched
 336 capacity. Relative to the matched global PINN, it reduces global concentration
 337 error by 9.1%, gross-power error by 9.4%, membrane-band concentration error
 338 by 20.3%, interface-flux imbalance by 31.7%, and mass-closure error by 19.1%.

339 FNO gives the lowest seed-averaged global concentration error, only 0.04
 340 percentage points below the membrane-aligned model. The membrane-aligned
 341 model gives lower errors for gross power, membrane-band concentration, flux
 342 imbalance, and mass closure. FNO efficiently interpolates the global solu-
 343 tion map from complete fields. The two-subdomain PINN applies water-
 344 permeation and solute-rejection conditions directly at the membrane, where
 345 averaged normal water flux enters the power calculation. The global concen-
 346 tration results are effectively tied; the membrane-local and power metrics
 347 separate the two models.

Table 3: Five-seed model comparison.

Model	Global concentration	Gross power	Membrane-band concentration	Flux imbalance	Mass closure
MLP-PINN	6.61 ± 0.17	5.66 ± 0.21	8.25 ± 0.35	3.13 ± 0.08	1.20 ± 0.08
Adaptive single ResNet-PINN	3.21 ± 0.36	3.00 ± 0.26	4.08 ± 0.27	1.98 ± 0.12	0.85 ± 0.07
Matched global ResNet-PINN	2.80 ± 0.43	2.65 ± 0.34	3.53 ± 0.34	1.64 ± 0.19	0.70 ± 0.05
DeepONet	2.79 ± 0.25	2.77 ± 0.25	3.92 ± 0.19	3.99 ± 0.05	3.03 ± 0.06
FNO	2.50 ± 0.47	2.49 ± 0.31	3.54 ± 0.35	3.62 ± 0.26	2.72 ± 0.05
POD-RBF	4.02 ± 0.28	3.76 ± 0.15	5.25 ± 0.19	5.05 ± 0.18	4.05 ± 0.07
Membrane-aligned ResNet-PINN	2.54 ± 0.17	2.40 ± 0.19	2.82 ± 0.11	1.12 ± 0.05	0.57 ± 0.07

Figure 2 compares the finite-element reference, MLP-PINN, and membrane-aligned model at the representative baseline condition. Differences are concentrated in the narrow membrane-adjacent region. Figure 3 resolves this region through centerline profiles; its inset enlarges the interface neighborhood, and “Full ResNet-PINN” denotes the membrane-aligned two-subdomain model.

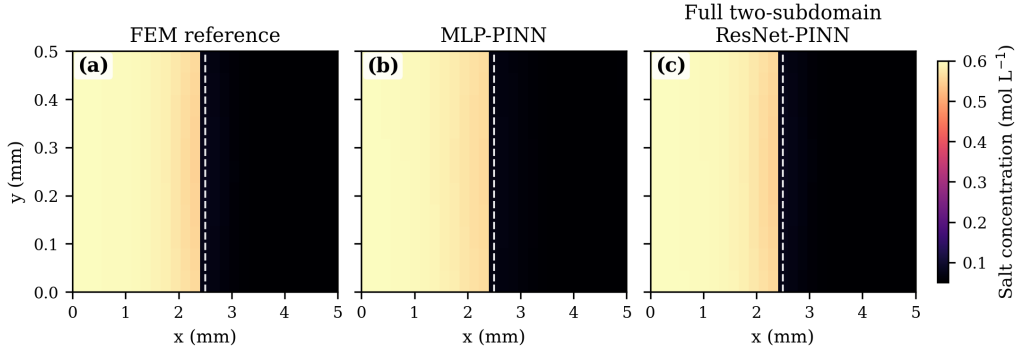


Figure 2: Concentration fields at the baseline condition.

All prespecified in-domain and OOD comparisons with the matched global PINN remain significant after Holm correction. For FNO, the in-domain global concentration difference is negligible ($r_{\text{rb}} = -0.032$, adjusted $p = 0.897$). Gross-power and interface metrics favor the membrane-aligned model, locating the main difference at the internal interface.

DeepONet, FNO, and POD-RBF use all nodes from 126 training fields. Each PINN receives anchors at 0.5% of the nodes in the same 126 cases, together with collocation residuals. The comparison spans complete-field operator learning and sparsely supervised equation-constrained learning. The membrane-aligned model reaches its reported power and interface errors with far fewer labeled nodal values. The finite-element fields still define the training cases and provide the final evaluation reference.

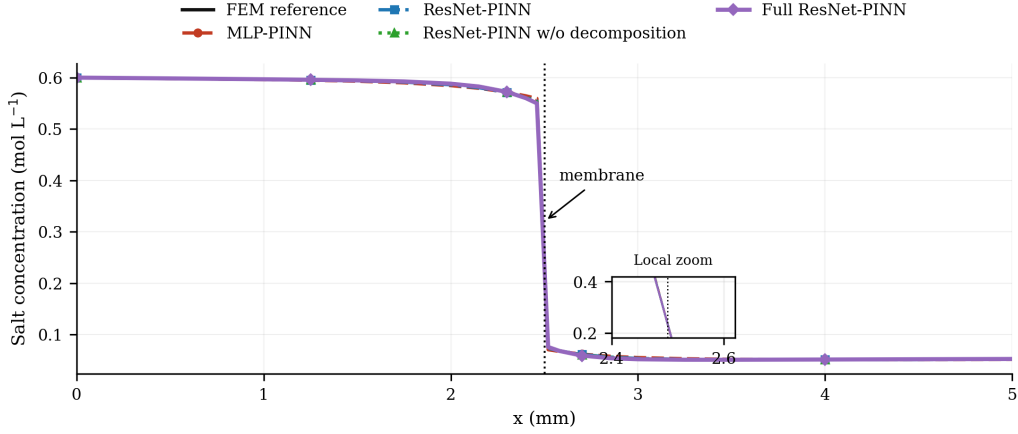


Figure 3: Centerline concentration profiles at the baseline condition.

Across the 27 in-domain tests, the membrane-aligned model has a median gross-power error of 2.40%, a 90th-percentile error of 2.77%, and a maximum of 2.86%. The corresponding concentration errors are 2.94%, 3.35%, and 3.99%. Its power error is lower than that of the matched global PINN on 70.4% of conditions and lower than FNO on 85.2%. The membrane-band win rates are 88.9% and 85.2%; interface-flux imbalance is lower on 96.3% and 100% of conditions. The advantage persists across the held-out operating set, with FNO remaining competitive in the global concentration norm.

5.2. Salinity-gradient power response and operating-window recovery

Pressure-power curves convert the predicted membrane fields into operating decisions for salinity-gradient power generation. Figure 4 shows representative slices through quadratic fits to the held-out results. The pressure slice uses $c_f = 0.04 \text{ mol L}^{-1}$ and $c_d = 0.70 \text{ mol L}^{-1}$; the salinity slice uses $\Delta P = 10 \text{ bar}$ and $c_f = 0.04 \text{ mol L}^{-1}$. These curves summarize the fitted surfaces and introduce no additional finite-element evaluations. The fitted finite-element response reflects the competition between hydraulic load and osmotic driving force. At fixed salinity, increasing pressure initially raises gross power through greater hydraulic work per unit water flux. At higher pressure, the smaller net osmotic driving force limits water transport. Higher draw salinity and lower feed salinity shift the response upward by increasing the available osmotic-pressure difference. The membrane-aligned fit follows the curvature and ordering of the finite-element fit. Accuracy near the peak

387 matters most for pressure selection. Vertical deviations on flat branches have
 388 little operational effect; a horizontal shift near the maximum changes the
 389 selected pressure.

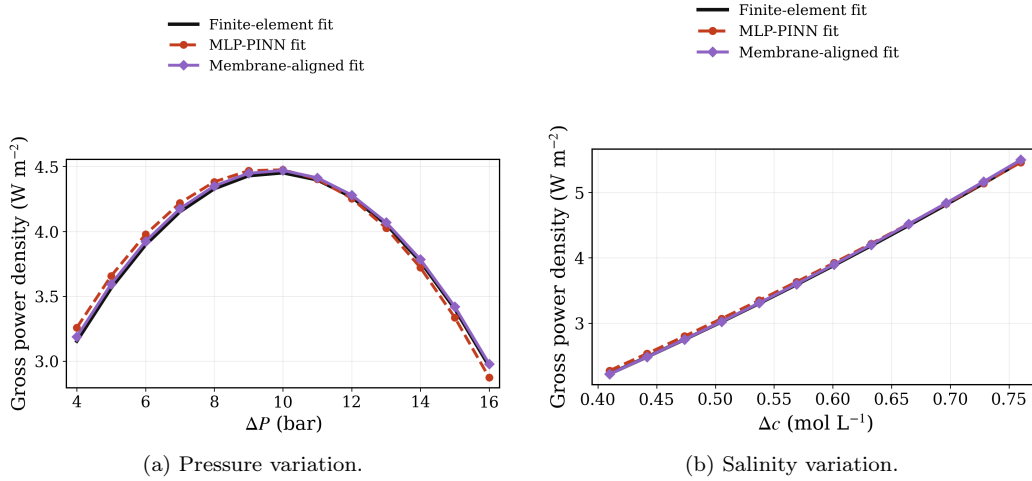


Figure 4: Gross-power response to pressure and salinity difference.

390 Five-fold cross-validation of the quadratic fits in Eq. (50) gives $R^2 =$
 391 0.960 for the finite-element output and $R^2 = 0.952$ for the surrogate output.
 392 The corresponding Root Mean Square Error (RMSE) values are 0.321 and
 393 0.355 W m⁻². These fits summarize the pressure–salinity response of each
 394 held-out data set. All neural-surrogate errors reported elsewhere use the
 395 original condition-wise predictions.

396 Figure 5 shows the fitted operating maps within the in-domain pressure
 397 and salinity bounds. The finite-element and surrogate surfaces are fitted
 398 independently; the plotted grid evaluates these fits and does not add reference
 399 simulations. Across the mapped feed and draw combinations, the finite-
 400 element surface places the optimum between approximately 6.85 and 11.37
 401 bar as salinity difference increases, with fitted maximum gross power between
 402 1.91 and 6.35 W m⁻². The optimal-pressure maps differ by 0.03 bar on average
 403 and by at most 0.083 bar. The maximum-power maps differ by 0.017 W m⁻²
 404 on average, equivalent to 0.48%. Applying the surrogate-selected pressure to
 405 the fitted finite-element surface gives a mean power regret of 0.0017% and a
 406 maximum of 0.0137%. Agreement is closest around the sampled optimum.
 407 Accuracy between sampled conditions is governed by the cross-validation
 408 error of the quadratic fits.

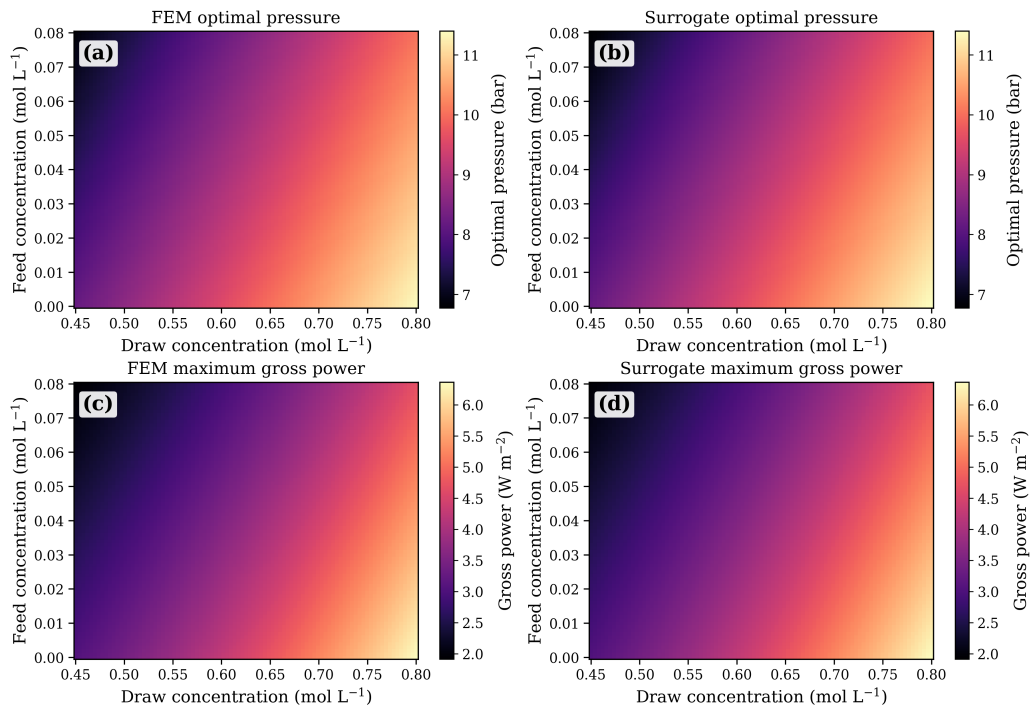


Figure 5: Operating-window maps from held-out cases.

Both fitted surfaces have a negative pressure-squared coefficient, producing a concave gross-power response across the sampled range. The positive pressure coefficient and positive pressure–salinity-gradient interaction move the maximizing pressure upward as $c_d - c_f$ increases. Feed salinity adds a negative contribution, consistent with the smaller osmotic driving force. The coefficient signs agree with the water-flux relation and the representative finite-element curves.

The 0.03-bar mean difference is small relative to the 12-bar pressure interval, and fitted-surface regret stays below 0.014% over the mapped salinity grid. The central salinity region is more strongly constrained than the corners. Finite-element confirmation is recommended for optima close to the pressure or salinity limits.

The rise in optimal pressure with salinity difference agrees with non-ideal process calculations [4]. Spiral-wound simulations show further shifts from flow arrangement and module hydraulics [2]; alternative layouts alter the balance between pressure loss and water transfer [3]. Full-scale optimization also finds a condition-dependent preferred pressure [1]. The maps resolve the membrane-scale pressure–salinity response and narrow the region requiring high-fidelity confirmation.

5.3. In-domain and boundary/OOD performance

Table 4 separates the in-domain and boundary/OOD partitions. All entries are mean condition-wise percentage errors. In-domain global concentration errors are similar for the membrane-aligned model and FNO. The membrane-aligned model gives lower gross-power and interface errors. On the boundary/OOD set, its mean power error is 3.888%, compared with 5.158% for the matched global PINN and 5.167% for FNO.

Table 4: Test and boundary/OOD errors.

Split	Model	Global concentration	Gross power	Membrane-band concentration	Flux imbalance
Test	Matched global ResNet-PINN	3.474	2.856	4.151	1.858
Test	FNO	2.891	2.915	4.005	3.744
Test	Membrane-aligned ResNet-PINN	2.904	2.418	3.337	1.306
OOD	Matched global ResNet-PINN	5.673	5.158	6.877	2.365
OOD	FNO	5.406	5.167	7.557	4.743
OOD	Membrane-aligned ResNet-PINN	4.586	3.888	4.912	1.594

Figure 6 separates the 30 boundary/OOD conditions into low-pressure, high-pressure/high-draw, high-feed/low-draw, and combined-extreme groups. Bars show category means, and error bars show one sample standard deviation.

Mean gross-power error ranges from 3.77% to 4.02%. Concentration error reaches its highest value for the combined extremes (5.39%), and interface-flux imbalance remains below 1.67% in every category. Power error is comparable across the tested boundary directions. Concentration error varies more and increases when a reduced salinity gradient coincides with multiple parameter excursions. The combined-extreme group contains two cases and is interpreted descriptively; the full OOD result uses all 30 conditions.

Across the OOD set, the membrane-aligned model has a median power error of 3.86%, a 90th-percentile error of 4.54%, and a maximum of 5.12%. Its maximum interface-flux imbalance is 1.99%. The largest relative power error occurs at high feed salinity and low draw salinity, where reference power is below 1 W m^{-2} and a modest absolute deviation yields a larger percentage. Maximum OOD power errors are 6.31% for the matched global PINN and 7.10% for FNO. The membrane-aligned model gives the lowest power error on 90.0% of conditions against the matched global PINN and 86.7% against FNO. It gives the lowest membrane-band and flux-imbalance errors on every OOD condition in both comparisons.

FNO performs well within the field distribution represented by the complete training snapshots. Its architecture carries no explicit membrane condition beyond that distribution. The global PINN retains the governing equations in a single continuous parameterization across both channels. The membrane-aligned model combines equation residuals with separate channel representations, and its membrane-local error grows more slowly under the tested excursions.

5.4. Sparse supervision, component contributions, and numerical consistency

Sparse finite-element anchors stabilize the parameterized physics-informed problem. With zero anchors, mean concentration, power, and membrane-band errors are 4.42%, 4.24%, and 5.52%, and one of five runs fails the validation threshold. At 0.5% of the nodes per training condition, all runs converge and the errors fall to 2.60%, 2.37%, and 2.98%. Anchor fractions above 1% produce little additional improvement in concentration. The selected 0.5% setting stabilizes the parameter-to-field map, with physics residuals retained throughout the domain.

Figure 7 shows a separate diagnostic run continued to 50,000 total iterations to display the Adam-to-L-BFGS transition and late-stage convergence. The vertical line marks the optimizer transition, and “Full ResNet-PINN”

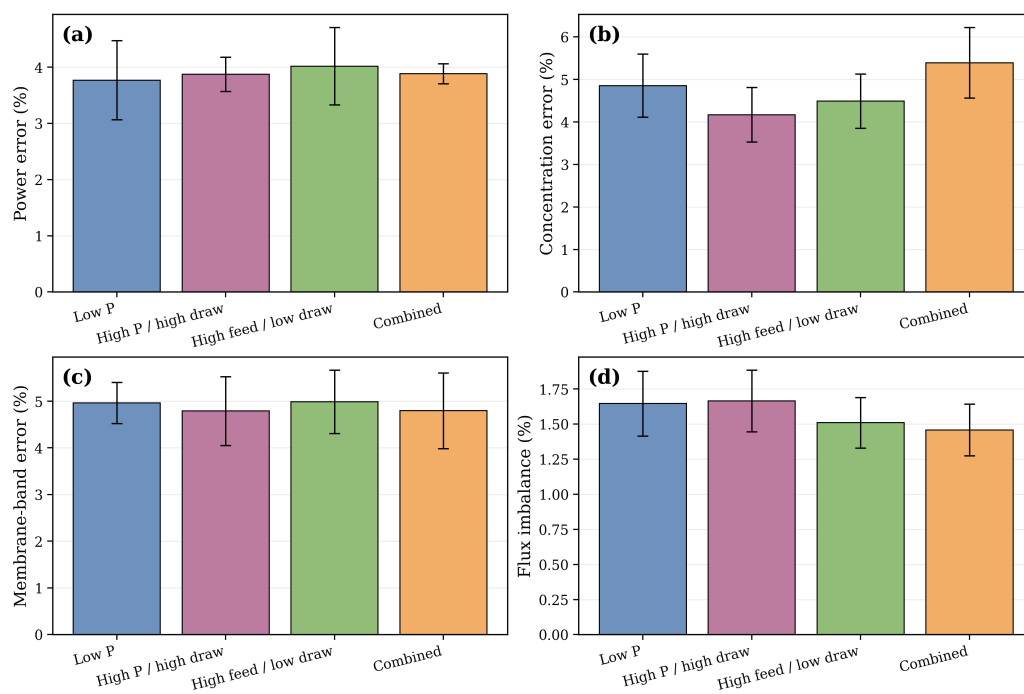


Figure 6: Boundary/OOD errors by category.

denotes the membrane-aligned model. Timing results use the validation-based early-stopped runs described above.

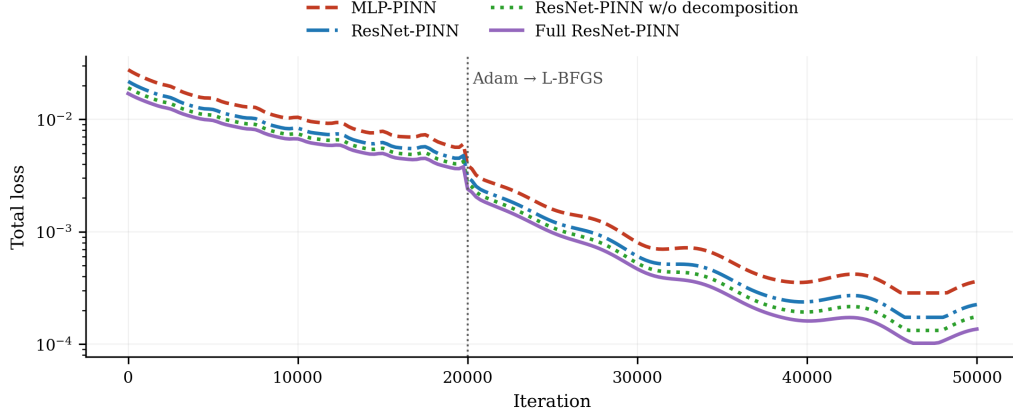


Figure 7: Optimization histories.

In the independently trained ablation batch, the interface loss has the largest effect. Removing the decomposition raises gross-power error from 2.52% to 2.79% and flux imbalance from 1.14% to 1.77%. Fixed loss weights raise power error to 3.47%; removing adaptive sampling raises it to 3.96%. Removing the explicit interface loss increases power, membrane-band, and flux-imbalance errors to 4.60%, 6.88%, and 5.94%. The architectural split reduces the approximation burden, and the membrane residuals govern water exchange and solute rejection between the two predictions. The anchor and ablation studies were trained as separate batches with their own validation checkpoints. Their full-model controls are compared within the corresponding batch. Complete five-seed records appear in the Supplementary Information.

The rounded summary in Fig. 8 reproduces the seed-level ordering. “Decomp” denotes removal of the membrane-aligned split, and the Supplementary Information reports the exact five-seed values. The largest degradation follows removal of the interface loss, followed by adaptive sampling and dynamic weighting.

Independent checks were performed on the finite-element references. Across six grid conditions, medium-to-fine differences average 0.371% for water flux, 0.415% for gross power, and 0.704% for membrane concentration. Across all 210 case records, mean COMSOL-FEniCS differences are 0.749% for velocity, 0.885% for pressure, 1.190% for concentration, and 1.516% for membrane flux.

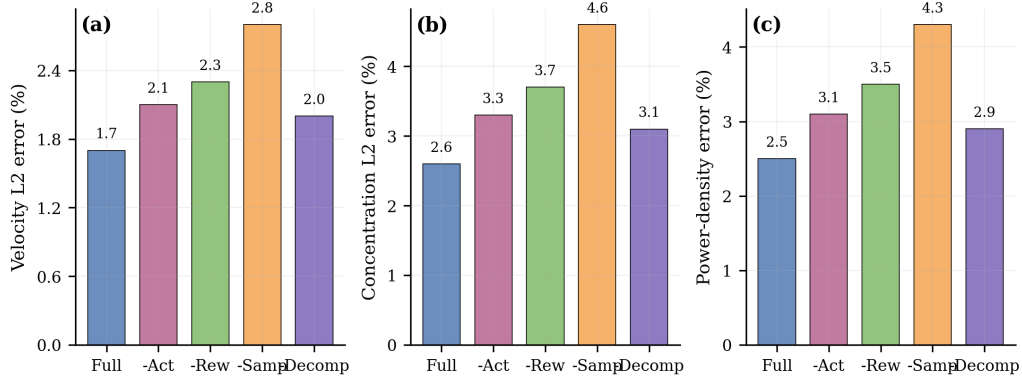


Figure 8: Ablation summary.

497 The two implementations agree within these ranges. Their concentration
 498 difference exceeds the 0.04-percentage-point gap between the seed-averaged
 499 global concentration errors of FNO and the membrane-aligned model. The
 500 global concentration results are treated as equivalent.

501 Figure 9 compares representative centerline profiles from COMSOL,
 502 FEniCS, the MLP-PINN, and the membrane-aligned model. The finite-
 503 element curves overlap closely. The membrane-aligned profile follows their
 504 membrane-adjacent transition more closely than the MLP-PINN.

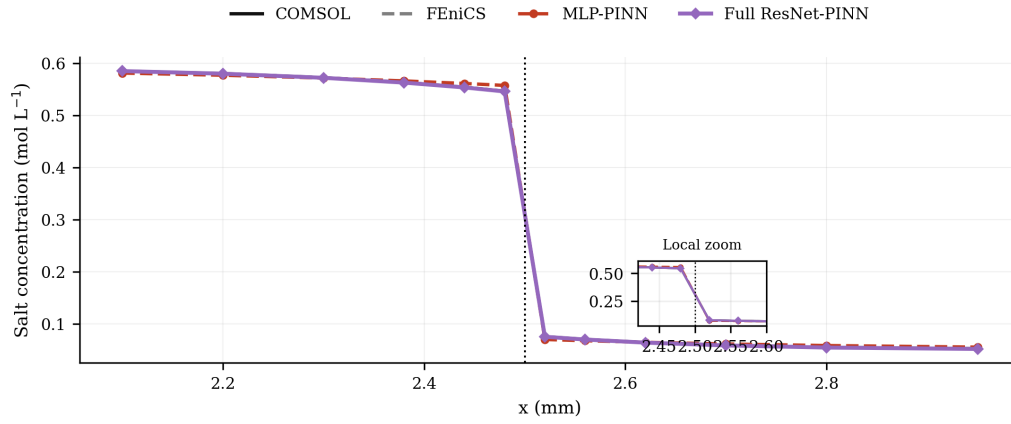


Figure 9: Centerline profiles across solvers and models.

505 *5.5. Computational value for salinity-gradient operating searches*

506 The mean finite-element solution time is 15.320 s per operating condition.
 507 Full-field inference with the membrane-aligned model takes 0.087 s, giving
 508 a $175\times$ online speedup. Including network training and generation of the
 509 126 training fields, end-to-end cost reaches break-even at approximately 504
 510 queries.

Table 5: Computational cost across query volumes.

Queries	Repeated FEM (h)	Surrogate end-to-end (h)	Time saving (%)
500	2.128	2.145	−0.8
1,000	4.256	2.157	49.3
10,000	42.555	2.375	94.4
100,000	425.553	4.561	98.9

511 Direct finite-element simulation remains competitive for a few hundred
 512 conditions because the surrogate carries an offline data-generation and training
 513 cost. Beyond the measured break-even point, savings rise quickly. End-to-end
 514 savings are 49.3% at 1,000 queries and 94.4% at 10,000 queries. Screening
 515 100,000 conditions would require an estimated 425.6 h with repeated finite-
 516 element solves and 4.56 h with the trained surrogate, including the same
 517 offline cost. These estimates use measured mean times and the common
 518 full-field output grid.

519 The trained model supports dense pressure scans for each feed–draw
 520 combination and rapid updates when the admissible salinity range changes.
 521 This is the main computational gain for salinity-gradient power studies, where
 522 many source and pressure combinations may need to be screened before
 523 detailed confirmation. The model can also provide sensitivity evaluations
 524 or initial populations for optimization without a finite-element solution for
 525 every candidate. POD–RBF and FNO have shorter inference times and larger
 526 membrane-local and power errors. The membrane-aligned model remains two
 527 orders of magnitude faster than the reference solver and returns the interface
 528 quantities used in the energy calculation.

529 *5.6. Implications for salinity-gradient power studies*

530 For each feed–draw salinity pair, the surrogate returns a pressure–power
 531 curve, a candidate operating pressure, and the membrane-side fields behind the

gross-power estimate. Finite-element solutions establish the initial pressure–salinity sample, after which the surrogate screens a denser set of conditions. Selected optima and boundary cases can then be returned to the finite-element solver. This sequence concentrates high-fidelity calculations near the operating region of interest.

The two-subdomain formulation mainly improves membrane-local quantities. Finite salt permeability introduces reverse salt flux and changes recoverable power [10]. Internal concentration polarization in the porous support changes membrane-side concentrations [39]. Channel geometry and pressure loss add further spatial variation [13]. These effects form part of detailed PRO models [14]. The aligned split keeps the feed and draw states distinct and places the corresponding constitutive laws at the membrane interface.

FNO gives the lowest seed-averaged global concentration error. The membrane-aligned PINN gives lower gross-power and interface errors. Reporting both global and membrane-local metrics reflects the quantities used in the power calculation.

The maps report membrane-scale gross power. Plant output also depends on membrane area and auxiliary equipment [4]. Channel-scale concentration and flow variations shape module hydraulics [11], and site conditions together with system-level operating choices determine recoverable output [1]. The surrogate accelerates the local transport calculations that feed into this broader analysis.

5.7. Scope and limitations

The power metric is gross areal power, $P = \bar{J}_w \Delta P$, and excludes pumping, pressure recovery, and auxiliary loads. The membrane is a zero-thickness interface with constant effective water permeability and complete salt rejection. The model excludes support-layer resistance, internal concentration polarization, reverse salt flux, spacers, fouling, concentration-dependent transport properties, and module-scale pressure loss. Osmotic pressure follows the van’t Hoff relation over the stated concentration range.

The operating maps apply to the membrane-scale model in Section 2 and are separate from commercial-module net output. Process-level analysis requires measured water and salt permeability coefficients [8], support-layer structural parameters and internal concentration polarization [39], and channel-scale hydraulic losses [13]. A full PRO model also requires non-ideal thermodynamics and validation against membrane data [14].

569 The quadratic surfaces summarize 57 held-out cases. Their cross-validation
 570 errors define the uncertainty between sampled conditions, and grid-to-grid
 571 differences compare two fitted summaries. The 30 boundary/OOD cases
 572 cover limited excursions around the training box, which defines the reported
 573 extrapolation range.

574 6. Conclusions

575 A membrane-aligned two-subdomain physics-informed surrogate was de-
 576 veloped for rapid operating-condition analysis of salinity-gradient power
 577 generation at membrane scale. It predicts the feed and draw transport fields,
 578 water flux, and gross power over changes in pressure and inlet salinity. Sepa-
 579 rate channel networks are coupled through water permeation, normal-velocity
 580 consistency, tangential no-slip, solute rejection, and electrical insulation.

581 Across five seeds, mean errors are 2.54% for concentration, 2.40% for gross
 582 power, 2.82% in the membrane band, and 1.12% for interface-flux imbalance.
 583 Capacity matching narrows the gain over a global PINN. The membrane-
 584 aligned model still gives lower power, membrane-band, flux-imbalance, and
 585 mass-closure errors. FNO is statistically equivalent in the in-domain global
 586 concentration norm and less accurate for the reported interface and power
 587 metrics.

588 Independent quadratic summaries of the held-out outputs place the fitted
 589 optimum between 6.85 and 11.37 bar over the modeled salinity range. The
 590 finite-element and surrogate optimal-pressure maps differ by 0.03 bar on
 591 average. Their maximum-power maps differ by 0.48%, and the mean fitted-
 592 surface power regret from the surrogate-selected pressure is 0.0017%.

593 Mean full-field inference is $175\times$ faster than the finite-element reference.
 594 Including training-field generation and network training, end-to-end cost
 595 breaks even after about 504 queries. Screening 10,000 conditions requires
 596 an estimated 42.6 h with repeated finite-element solves and 2.38 h with the
 597 surrogate. The model screens pressure and salinity combinations within
 598 the tested membrane-scale range. Extension to process scale requires non-
 599 ideal membrane transport, module hydraulics, net-power accounting, and
 600 experimental validation.

601 Data availability

602 The data supporting the findings of this study are available from the
 603 authors upon reasonable request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of Competing Interest

The authors declare no competing financial interests or personal relationships that could have influenced the work reported in this paper.